Smart Brewery

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BEER THAT MAKES YOU SMART

Features

- Brewing beer according to predefined recipes: a smart box connected to a brewing system that can monitor temperature and control heating elements.
- Application: control and brew through a Web-Application, allowing automatic brewing and sending the user notifications when needed human intervention.
- Recipe sharing: the option to download a recipe and load it into the smart box in a "download and brew" mode using the computer's automatic control of the brewing systems sensors.
- Research: monitoring measurements in the beer preparation process and analyzing them by AI to identify relationships and effects on the brewing outcome.

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Features

- Fermentation Monitoring: Include sensors to monitor the fermentation process, tracking factors like humidity, and temperature. Provide alerts or suggestions for adjustments if needed.

- Batch History and Analytics: Keep a history of each brewing batch, including all relevant data. Provide analytics and insights to help users refine their brewing techniques over time.

- Community and Social Features: Build a platform where users can share their recipes, brewing experiences, and tips. This could include a social feed within the app.

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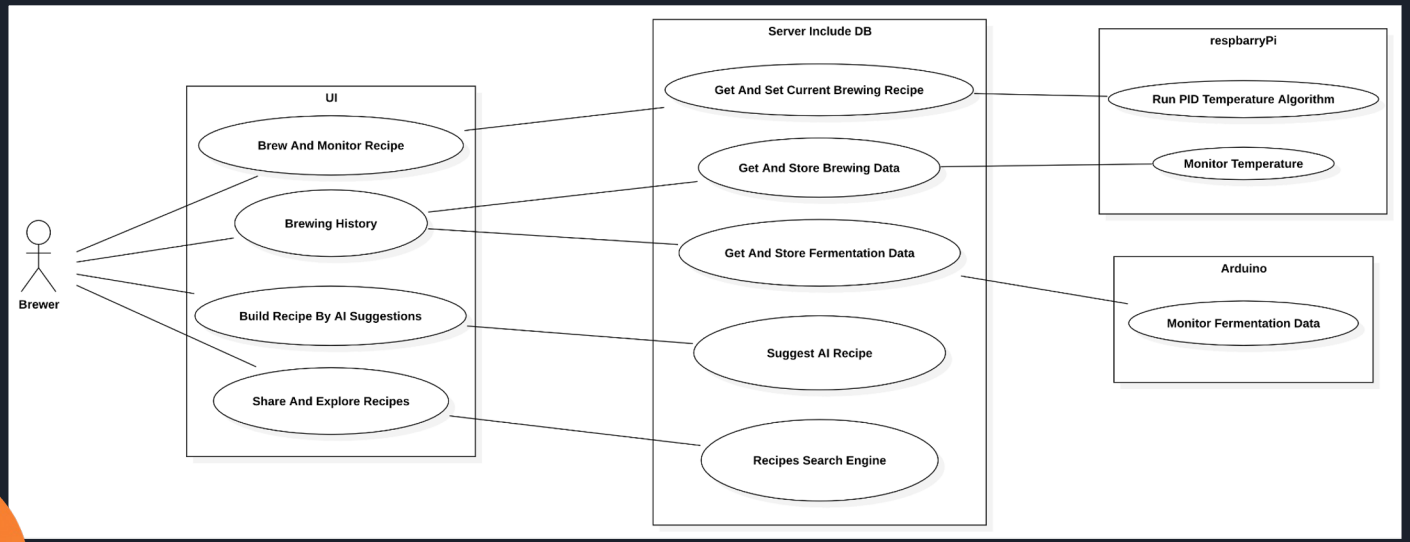
Architecture

Use-Case Diagram



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Embedded

Development languages : Python, C++

FrameWorks : RaspberryPi Os, Arduino IDE

Internet Connectivity :wifi

Api Communication :HTTP microservices

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Back-End

Development language : Java

FrameWork : Maven, Spring Boot

Data Base : PostgreSQL

Api Communication :HTTP microservices

Front-End

Development language :JS

FrameWork : React

Application : Web

Api Communication :HTTP